

10 - Storage Database Requirements

Visor DOM-to-Agent Context Compiler
DOC-10 | 0.1 requirements pack | Generated 2026-06-14

This specification defines local extension storage, data retention policies, future database model, migrations, and do-not-store rules.

Field	Value
Owner	Rohan PX / Visor project
Audience	AI coding agent, reviewer, future contributors
Status	Draft baseline for MVP build
Decision rule	MVP is local-first, minimal-permission, Chrome Manifest V3, no backend unless explicitly enabled.

Document control

Item	Value
Project	Visor DOM-to-Agent Context Compiler
Document code	DOC-10
Generated on	2026-06-14
Primary goal	Turn raw webpage DOM into safe, structured, agent-ready context.
MVP boundary	Chrome extension, local-only compiler, JSON/Markdown export, privacy redaction, tests.

1. Storage principle

The MVP stores settings and site profiles locally. It should not store full page contents by default. Full compiled contexts may exist temporarily in memory for preview and clipboard export.

2. Local storage objects

```
type UserSettings = {
  defaultMode: 'compact' | 'detailed' | 'agent_action' | 'rag' | 'debug';
  privacyLevel: 'low' | 'medium' | 'strict';
  tokenBudget: number;
  defaultExport: 'json' | 'markdown' | 'prompt_block';
  debugMode: boolean;
  blockedDomains: string[];
};

type SiteProfile = {
  id: string;
  domain: string;
  preserveSelectors: string[];
  ignoreSelectors: string[];
  mainContentSelector?: string;
  privacyLevelOverride?: 'low' | 'medium' | 'strict';
  createdAt: string;
  updatedAt: string;
};

type RecentCompileMetadata = {
  id: string;
  url: string;
  title: string;
  createdAt: string;
  mode: string;
  tokenCount: number;
  riskLevel: string;
};
```

3. Storage policy

Object	Storage location	Retention	Notes
UserSettings	chrome.storage.local or browser storage adapter	Until user clears	Small and non-sensitive.
SiteProfile	Local extension storage	Until deleted	Domain rules may reveal habits; allow export/delete.
RecentCompileMetadata	Optional local storage	Configurable or off	Do not store full content.
Full PageSnapshot	Memory only in MVP	Until popup/preview closes	No persistence by default.
Full AgentContext	Memory/clipboard/download by user action	User-controlled	No automatic storage.

4. Database future model

```

users(id, email_hash, created_at, settings_json)
site_profiles(id, user_id, domain, preserve_selectors_json, ignore_selectors_json,
  created_at, updated_at)
contexts(id, user_id, title, url_hash, url_encrypted, context_encrypted, token_count,
  risk_level, created_at)
audit_events(id, user_id, event_type, object_id, created_at, metadata_json)
api_keys(id, user_id, key_hash, scopes_json, created_at, revoked_at)

```

5. Migration requirements

ID	Requirement	Priority	Acceptance signal
DB-001	Every local object must include schemaVersion or appVersion metadata.	P0	Migration tests can detect old settings.
DB-002	Storage adapter must supply defaults when stored data is missing or invalid.	P0	Invalid settings fixture falls back safely.
DB-003	User must be able to clear local settings and metadata.	P0	Clear local data button works.
DB-004	Site profiles must be exportable/importable as JSON.	P1	User can move rules across browsers.
DB-005	Future cloud storage must support user deletion.	P0 future	Deletion workflow is documented before backend launch.

6. Do-not-store list

- Password field values.
- OTP values.
- Session tokens or cookies.
- Full private dashboard content by default.
- Banking, medical, government, or educational portal content without explicit save action.
- Authentication headers or browser cookies.

Source and reference notes

These sources are used to ground platform/security assumptions. The implementation must still verify current platform behavior during development.

Reference	Why it matters
MDN WebExtensions - storage API: https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/API/storage	Grounding for extension storage concepts and permissions.